an indeterminate order. Accordingly, you learn about testing in synchronous and asynchronous settings.

4.1 Testing approaches

To test actors—if they are written in Scala—you work with the ScalaTest framework. ScalaTest is an xUnit-style testing framework designed for readability, so it should be easy to read and understand the intent of the test. With this framework alone, testing actors can pose some difficulties:

- *Expected order*—Sending messages is asynchronous in a real scenario, which makes it difficult to specify the order in which the expected values should be asserted in the unit test.
- *Statelessness*—An actor hides its internal state and does not allow direct access to it. Access should be possible only through the `ActorRef`. Sending messages to an actor is our only way to know its internal state (which is limited to the state the actor had when it replied).
- *Collaboration/integration*—If you want to do an integration test for multiple actors, you need to eavesdrop on the actors to assert that the messages have the expected values. It is not immediately clear how to do this. You'll learn to do this with a probe.

Fortunately, Akka has the `akka-testkit` module, which contains a set of testing tools that make actor testing much easier. This module enables several types of testing: